

# A FORMALLY VERIFIED SPARSE FOCK CONSTRUCTION FOR ERDŐS PROBLEM 906

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ABSTRACT. Consider the explicit sparse Fock series

$$F(z) = \sum_{j \geq 1} \frac{z^{\lfloor j^{3/2} \rfloor}}{\sqrt{\lfloor j^{3/2} \rfloor}!}.$$

For every nonempty open set  $U \subset \mathbb{C}$ , the derivative  $F^{(n)}$  has a zero in  $U$  for every sufficiently large  $n$ . On each fixed annulus, the zeros have a quantitative two-term description: every zero lies in a model disk, the relevant model disks are eventually pairwise disjoint, each contains exactly one simple zero, and the covering radius is  $O(n^{-1/6})$ . The construction is deterministic. The accompanying Lean 4/Mathlib development verifies every theorem, proposition, and corollary stated in this note and formalizes the explicit inequalities used in their proofs.

## 1. STATEMENT AND CONTEXT

The literal formulation of Erdős Problem 906 allows a polynomial, since all sufficiently high derivatives then vanish identically. The nontrivial form asks for a transcendental entire function. For such a function  $g$ , write  $\mathcal{Z}(g)$  for its zero set and consider the cofinite zero-hitting property

$$\forall U \subset \mathbb{C} \text{ nonempty open,} \quad \exists N(U) \quad \forall n \geq N(U) : \quad \mathcal{Z}(g^{(n)}) \cap U \neq \emptyset. \quad (1.1)$$

This has the same quantifier content as the usual formulation using every strictly increasing sequence of derivative orders. If (1.1) fails for some  $U$ , the infinitely many exceptional orders contain a strictly increasing sequence all of whose selected derivatives miss  $U$ . Conversely, if (1.1) holds, then every strictly increasing sequence  $n_k$  eventually satisfies  $n_k \geq N(U)$ , so all sufficiently late selected derivatives have a zero in  $U$ .

Erdős recorded the problem in [2, p. 72] and wrote that the two existence questions stated there had been proved more than ten years earlier, but he supplied no proof or precise reference for the cofinite statement. Barth and Schneider's 1972 theorem [1] proves a related but different interpolation statement: for prescribed discrete sets  $S_k$ , one may choose derivative orders  $n_k$  and a transcendental entire function whose  $n_k$ -th derivative vanishes on  $S_k$ . Their result does not contain the cofinite quantifier in (1.1). Gethner's final-set theory likewise concerns points approached by zeros of infinitely many derivatives, again a weaker quantifier than (1.1) [3]. Hou gave a probabilistic bounded-coefficient Fock-series construction satisfying (1.1), together with a Lean formalization of the main theorem and growth estimate [4]. The present construction is explicit and sparse, and its derivative zeros admit a quantitative localization theory. I make no historical-priority claim beyond these stated features.

Put

$$\nu_j = \lfloor j^{3/2} \rfloor, \quad q_j = \nu_{j+1} - \nu_j, \quad F(z) = \sum_{j=1}^{\infty} \frac{z^{\nu_j}}{\sqrt{\nu_j}!}. \quad (1.2)$$

For  $0 < a \leq b$ , let

$$A_{a,b} = \{z \in \mathbb{C} : a \leq |z| \leq b\}.$$

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For an active index  $j$ , meaning  $\nu_j \geq n$ , define

$$m_j = \nu_j - n, \quad A_j = \frac{\sqrt{\nu_j!}}{m_j!}, \quad T_j(z) = A_j z^{m_j}, \quad \rho_j = \left( \frac{A_j}{A_{j+1}} \right)^{1/q_j}. \quad (1.3)$$

Thus  $|T_j| = |T_{j+1}|$  on  $|z| = \rho_j$ . The nonzero zeros of the two-term model  $T_j + T_{j+1}$  are

$$z_{j,\ell} = \rho_j \exp\left(\frac{(2\ell+1)\pi i}{q_j}\right), \quad 0 \leq \ell < q_j. \quad (1.4)$$

For  $c > 0$ , write

$$\mathcal{D}_{j,\ell}(c) = \overline{D}\left(z_{j,\ell}, \frac{c\rho_j}{q_j}\right). \quad (1.5)$$

**Theorem 1.1** (Cofinite derivative zeros). *For every nonempty open set  $U \subset \mathbb{C}$ , there is  $N$  such that for every  $n \geq N$ , the derivative  $F^{(n)}$  has a zero in  $U$ .*

**Theorem 1.2** (Quantitative annular covering). *Let  $0 < a \leq b$ . There are constants  $C > 0$  and  $N$  such that, for every  $n \geq N$  and every  $w \in A_{a,b}$ , there is a zero  $z$  of  $F^{(n)}$  satisfying*

$$|z - w| \leq Cn^{-1/6}. \quad (1.6)$$

**Proposition 1.3** (One simple zero in each model disk). *Let  $0 < a \leq b$ . There exist  $c \in (0, 1]$  and  $N$  such that, for every  $n \geq N$ , every active  $j$  with  $a \leq \rho_j \leq b$ , and every  $0 \leq \ell < q_j$ , the disk  $\mathcal{D}_{j,\ell}(c)$  contains exactly one zero of  $F^{(n)}$ , and that zero is simple.*

**Proposition 1.4** (Exhaustion of annular zeros). *Let  $0 < a \leq b$  and  $c > 0$ . For all sufficiently large  $n$ , every zero  $z$  of  $F^{(n)}$  with  $a \leq |z| \leq b$  belongs to a disk  $\mathcal{D}_{j,\ell}(c)$  with*

$$\frac{a}{2} \leq \rho_j \leq 2b, \quad 0 \leq \ell < q_j.$$

**Corollary 1.5** (Eventual annular simplicity). *Let  $0 < a \leq b$ . For all sufficiently large  $n$ , every zero of  $F^{(n)}$  in  $A_{a,b}$  is simple.*

**Proposition 1.6** (Pairwise disjoint model disks). *Let  $0 < a \leq b$  and  $0 < c \leq 1$ . For all sufficiently large  $n$ , the disks  $\mathcal{D}_{j,\ell}(c)$  with active  $j$ ,  $a \leq \rho_j \leq b$ , and  $0 \leq \ell < q_j$  are pairwise disjoint.*

**Proposition 1.7** (Finite model-disk zero count). *Let  $0 < a \leq b$ . There exist  $c \in (0, 1]$  and  $N$  such that, for every  $n \geq N$  and every finite set  $I$  of pairs  $(j, \ell)$  with active  $j$ ,  $a \leq \rho_j \leq b$ , and  $0 \leq \ell < q_j$ , the union of the disks indexed by  $I$  contains exactly  $|I|$  distinct zeros of  $F^{(n)}$ , all simple.*

The rest of the paper proves these statements in the same structural order as the formal development. Results for other support exponents, the endpoint  $p = 4/3$ , sectorial asymptotics, limiting zero measures, and exact order and type are not part of this submission.

## 2. SUPPORT, DERIVATIVE WEIGHTS, AND CROSSING RADII

The support has one relevant arithmetic scale. By the mean-value theorem and the floor error,

$$\frac{3}{2}\sqrt{j} - 1 \leq q_j \leq \frac{3}{2}(\sqrt{j} + 1) + 1 \quad (j \geq 1). \quad (2.1)$$

Hence, whenever  $\nu_j = n + O(\sqrt{n})$ ,

$$j \asymp n^{2/3}, \quad q_j \asymp n^{1/3}. \quad (2.2)$$

In particular, if  $j_0 = \min\{j : \nu_j \geq n\}$ , then  $m_{j_0} = O(n^{1/3})$ .

The series (1.2) is entire. A uniform majorant, proved in the formal development by Cauchy–Schwarz, is

$$|F(z)| \leq \sqrt{1 + |z|^2} \exp\left(\frac{1 + |z|^2}{2}\right). \quad (2.3)$$

Termwise differentiation is therefore legitimate, and

$$F^{(n)}(z) = \sum_{\nu_j \geq n} A_j z^{m_j}. \quad (2.4)$$

The support is infinite, so  $F$  is not a polynomial. The formal proof of transcendence is slightly stronger than this observation: a sufficiently high derivative of a polynomial vanishes identically, whereas the corresponding derivative of  $F$  does not.

For integers  $k \geq n$  and  $r > 0$ , set

$$\Phi_{n,r}(k) = \frac{1}{2} \log(k!) - \log((k-n)!) + (k-n) \log r. \quad (2.5)$$

Its first difference is

$$d_{n,r}(k) = \Phi_{n,r}(k+1) - \Phi_{n,r}(k) = \log r + \frac{1}{2} \log(k+1) - \log(k+1-n). \quad (2.6)$$

Writing  $m = k - n$  gives

$$d_{n,r}(k) - d_{n,r}(k+1) = \log\left(1 + \frac{1}{m+1}\right) - \frac{1}{2} \log\left(1 + \frac{1}{n+m+1}\right) > 0. \quad (2.7)$$

Thus  $\Phi_{n,r}$  is strictly discretely concave. Uniformly for  $m = O(\sqrt{n})$ , the drop in (2.7) is comparable to  $n^{-1/2}$ .

The crossing radius has the exact representation

$$\log \rho_j = \frac{1}{q_j} \sum_{s=1}^{q_j} \left\{ \log(m_j + s) - \frac{1}{2} \log(n + m_j + s) \right\}. \quad (2.8)$$

The summand is increasing in  $s$ , hence the crossing radii are strictly increasing. The same averaging formula gives the explicit squeeze

$$\frac{m_j + 1}{\sqrt{n + m_j + 1}} \leq \rho_j \leq \frac{m_j + q_j}{\sqrt{n + m_j + q_j}}. \quad (2.9)$$

In the saddle range  $m_j = O(\sqrt{n})$ , (2.9) and (2.2) give

$$\rho_j = \frac{m_j}{\sqrt{n}} + O(n^{-1/6}). \quad (2.10)$$

On every fixed compact annulus, consecutive relevant crossing radii satisfy

$$c_{a,b} n^{-1/6} \leq \rho_{j+1} - \rho_j \leq C_{a,b} n^{-1/6} \quad (2.11)$$

for all sufficiently large  $n$ , after enlarging the annulus by a fixed factor. The radial mesh is therefore  $n^{-1/6}$ , whereas the angular mesh on a crossing circle is  $q_j^{-1} = O(n^{-1/3})$ .

### 3. TAIL DOMINATION AND LOCAL ZERO TRANSFER

Fix a compact saddle range and a relevant crossing  $\rho = \rho_j$ . Put

$$m = m_j, \quad q = q_j, \quad Q = \min(q_{j-1}, q_{j+1}).$$

For large  $n$  the index  $j$  is at least 2. The curvature estimate provides a positive number  $c$  such that the full-weight slope at the left end of the block is at least  $c$  and at the right end is at most  $-c$ . In the present  $p = 3/2$  situation,

$$c \asymp n^{-1/6}, \quad Q \asymp n^{1/3}. \quad (3.1)$$

Let  $0 \leq \gamma < c$  and write  $|z| = \rho e^\sigma$  with  $|\sigma| \leq \gamma$ . The formal tail estimate is

$$\frac{\sum_{h \neq j, j+1} |T_h(z)|}{|T_j(z)|} \leq e^{-(c-\gamma)Q} \left( (m+1) + \frac{e^{q\gamma}}{1 - e^{-(c-\gamma)}} \right). \quad (3.2)$$

This is the estimate used below; in particular, no geometric-series simplification of the left tail is needed. If  $\gamma = C_0/q$  with fixed  $C_0 > 0$ , then  $q\gamma = O(1)$ ,  $(c-\gamma)Q \asymp n^{1/6}$ , and  $m = O(\sqrt{n})$ . Thus the right-hand side of (3.2) tends to zero despite the polynomial factor. The explicit numerical lemmas in the Lean development choose a threshold  $N$  at which it is smaller than each constant required in the local arguments.

We now prove Proposition 1.3. Let  $z_0 = z_{j,\ell}$  and  $q = q_j$ . In logarithmic coordinates  $z = z_0 e^w$ , divide by the nonvanishing principal monomial. The derivative takes the form

$$\Psi(w) = 1 - e^{qw} + E(w). \quad (3.3)$$

Choose once and for all  $0 < c_0 \leq 1/16$  and put  $\zeta = c_0/q$ . Applying (3.2) on  $|w| \leq \zeta$  gives  $|E(w)| \leq K_n$ , with  $K_n \rightarrow 0$ . At  $w = 0$ ,  $|\Psi(0)| \leq K_n$ . On  $|w| = \zeta$ , the elementary exponential remainder estimate gives

$$|1 - e^{qw}| \geq c_0/2. \quad (3.4)$$

For large  $n$ ,  $K_n < c_0/4$ , so  $|\Psi| > K_n$  on the boundary. If  $\Psi$  had no zero in the closed disk, the maximum-modulus principle applied to  $1/\Psi$  would contradict the smaller value of  $|\Psi|$  at the centre. Hence there is a zero  $z$  with

$$|z - z_0| \leq 2\rho\zeta = \frac{2c_0\rho}{q}. \quad (3.5)$$

This is the minimum-modulus step formalized in `TwoTerm.lean`.

For uniqueness and simplicity, set  $t = z/z_0$  and normalize again:

$$G(t) = 1 - t^q + R(t). \quad (3.6)$$

Take  $s = 2c_0/q$ , so  $qs = 2c_0 \leq 1/8$ , and use the tail estimate on the doubled disk  $|t - 1| \leq 2s$ . The logarithmic radial shift there is at most  $4s$ , so the same tail theorem applies after increasing the harmless fixed transition constant. After replacing  $K_n$  by this larger tail bound, we have  $|R| \leq K_n$  on  $|t - 1| \leq 2s$ , and Cauchy's estimate gives

$$|R'(t)| \leq K_n/s \quad (|t - 1| \leq s). \quad (3.7)$$

On this disk,  $|t^{q-1} - 1| \leq 1/7$  for the fixed choice above, and therefore

$$|t_1^q - t_2^q| \geq \frac{6q}{7} |t_1 - t_2| \quad (|t_i - 1| \leq s). \quad (3.8)$$

For large  $n$ ,  $K_n < qs/2$ . If  $G(t_1) = G(t_2) = 0$ , (3.7) and (3.8) force  $t_1 = t_2$ . At the zero,

$$|qt^{q-1}| \geq 6q/7 > K_n/s,$$

so  $G'(t) \neq 0$ . The zero from (3.5) lies in this uniqueness disk, proving Proposition 1.3 with the common disk constant  $c = 2c_0$  after increasing the threshold  $N$  if necessary.

#### 4. EXHAUSTION, DISJOINTNESS, AND COVERING

Fix  $r$  in a compact annulus. Because the crossing radii are strictly increasing, for all large  $n$  there is a crossing index  $j$  with

$$\rho_j \leq r < \rho_{j+1}. \quad (4.1)$$

Put

$$\sigma = \log(r/\rho_j) \geq 0, \quad \tau = \log(\rho_{j+1}/r) \geq 0.$$

The formal exclusion argument has exactly three cases. For a fixed sufficiently small  $c_0 > 0$ :

- (1) if  $\sigma \leq c_0/q_j$ , the two-term transition estimate at  $\rho_j$  puts every zero on  $|z| = r$  in a model disk attached to  $\rho_j$ ;
- (2) if  $\tau \leq c_0/q_{j+1}$ , the same argument at  $\rho_{j+1}$  puts every zero in a model disk attached to  $\rho_{j+1}$ ;

- (3) if both inequalities fail, the middle term  $T_{j+1}$  is larger than the sum of all the remaining terms, so  $F^{(n)}$  has no zero on that circle.

The third case follows from the same concavity and tail estimate as above, now applied on both sides of the dominant term. In the first two cases, the transition width is eventually below  $1/2$  in logarithmic radius; hence a zero with  $a \leq |z| \leq b$  can only be attached to a crossing radius in  $[a/2, 2b]$ . This proves Proposition 1.4.

Apply Proposition 1.3 on  $A_{a/2, 2b}$  and use its disk constant in Proposition 1.4. Every annular zero lies in a disk whose unique zero is simple. Corollary 1.5 follows.

For Proposition 1.6, two distinct model centres on the same crossing circle satisfy

$$|z_{j,\ell} - z_{j,\ell'}| \geq \frac{4\rho_j}{q_j} \quad (\ell \neq \ell') \quad (4.2)$$

for all sufficiently large relevant  $j$ . Hence disks with  $c \leq 1$  on the same circle are disjoint. On different circles, (2.11) gives radial separation of order  $n^{-1/6}$ , whereas the disk radii are  $O(q_j^{-1}) = O(n^{-1/3})$ . Thus the whole admissible family is eventually pairwise disjoint. Proposition 1.7 then follows by combining disjointness with the one-simple-zero statement in each disk.

We finally prove Theorem 1.2. Given  $w \in A_{a,b}$ , set  $r = |w|$  and choose the crossing index from (4.1). The upper bound in (2.11) gives

$$|r - \rho_j| \leq C_{a,b} n^{-1/6}. \quad (4.3)$$

Choose the nearest model angle on  $|z| = \rho_j$ . Its angular displacement from  $w$  is  $O(q_j^{-1}) = O(n^{-1/3})$ . Proposition 1.3, applied on a fixed enlargement of the annulus, supplies a true zero within another  $O(\rho_j/q_j) = O(n^{-1/3})$  of that model point. The radial error (4.3) dominates, proving (1.6).

Now let  $U \subset \mathbb{C}$  be nonempty and open. Choose a nonzero  $w \in U$  and a closed disk  $\overline{D}(w, \delta) \subset U$  with  $\delta < |w|/2$ . This disk is contained in a fixed annulus away from the origin. For all sufficiently large  $n$ , the covering radius in Theorem 1.2 is smaller than  $\delta$ , so  $F^{(n)}$  has a zero in  $U$ . This proves Theorem 1.1.

*Remark 4.1* (The origin). Let  $j_0 = \min\{j : \nu_j \geq n\}$ . The multiplicity of the zero of  $F^{(n)}$  at the origin is exactly  $\nu_{j_0} - n$ . This can be large, so the simplicity statement is necessarily restricted to annuli bounded away from the origin.

## 5. FORMAL VERIFICATION AND SOURCE

The Lean development uses Lean 4.28.0 and the Mathlib revision pinned by the project manifest. It is available with the paper at

<https://github.com/FDmd233/lean4-math-showcase/tree/main/projects/erdos-906-sparse-fock>.

The canonical entry point is `formalization/lean/RequestProject/Main.lean`. It imports the complete proof chain and prints the axiom dependencies of every declaration in the paper-to-Lean crosswalk. The main correspondences are:

| Printed statement                      | Lean declaration                                             |
|----------------------------------------|--------------------------------------------------------------|
| Entire function and derivative formula | <code>F_differentiable</code> , <code>iteratedDeriv_F</code> |
| Transcendence                          | <code>F_not_polynomial</code>                                |
| Upper growth bound (2.3)               | <code>F_norm_le</code>                                       |
| Theorem 1.2                            | <code>annular_covering_rate_p32</code>                       |
| Theorem 1.1                            | <code>erdos906_sparse_fock_p32</code>                        |
| Proposition 1.3                        | <code>model_disk_zero_count_one_p32</code>                   |
| Proposition 1.4                        | <code>annular_zero_exclusion_p32</code>                      |
| Corollary 1.5                          | <code>annular_zeros_simple_deriv_p32</code>                  |
| Proposition 1.6                        | <code>model_disks_pairwise_disjoint_p32</code>               |
| Proposition 1.7                        | <code>zero_count_model_disk_union_p32</code>                 |
| Remark 4.1                             | <code>origin_multiplicity</code>                             |

The exact tail inequality (3.2) is `tail_sum_bound` in `Tail.lean`; the minimum-modulus localization is formalized in `TwoTerm.lean`; the local uniqueness and simplicity estimate is formalized in `ZeroCount32.lean`; and the three-case exclusion argument is formalized in `Exclusion.lean` and `Annulus.lean`. The repository contains a more detailed proof-to-code map.

The checked project modules pass the repository’s static proof-integrity policy before compilation. The declarations printed by `Main.lean` use only the standard dependencies `propext`, `Classical.choice`, and `Quot.sound`. Continuous integration then builds the canonical entry point from the pinned Lean toolchain.

#### AI ASSISTANCE DISCLOSURE

Language-model tools were used substantially during derivation, checking, exposition, repository maintenance, and formalization. The repository separates the machine-checked submission from research material that is not part of this paper.

#### REFERENCES

- [1] K. F. Barth and W. J. Schneider, *On a problem of Erdős concerning the zeros of the derivatives of an entire function*, Proc. Amer. Math. Soc. **32** (1972), 229–232. DOI: 10.2307/2038336.
- [2] P. Erdős, *Some of my favourite problems which recently have been solved*, in L. H. Y. Chen, T. B. Ng and M. J. Wicks (eds.), Proceedings of the International Mathematical Conference, Singapore 1981, North-Holland Math. Stud. **74** (1982), 59–79. DOI: 10.1016/S0304-0208(08)70415-8.
- [3] R. M. Gethner, *On the zeros of the derivatives of some entire functions of finite order*, Proc. Edinburgh Math. Soc. **28** (1985), 381–407. DOI: 10.1017/S001309150001720X.
- [4] E. Hou, *Cofinite Zeros of High Derivatives*, arXiv:2607.20816 (2026). arXiv:2607.20816.